Chapter 16 — Action and Transaction Architecture

Field	Value
Status	Source-aligned rewrite; architecture audit passed
Version	0.2
Scope	Turn approved intent into recoverable, observable execution with explicit commit barriers, rollback and compensation.
Constitutional basis	Decisions 9, 16, 20–22, 25, 29, 35, 40, 50, 55–56
Owner review	None required

1. Purpose

Turn approved intent into recoverable, observable execution with explicit commit barriers, rollback and compensation.

2. Authoritative Responsibilities

Component	Authoritative responsibility
Transaction Coordinator	Transaction state, journals, commit barriers and recovery
Execution Orchestrator	Provider/node binding, ordering, concurrency, retries and cancellation
Effect Registry	Declared side effects, reversibility, idempotency and compensation
Verification Coordinator	Precondition, dry-run and outcome verification

3. Core State and Records

Object	Required content
Transaction	Intent, plan, actor, authority, policy, state,

Object	Required content
	effects, recovery and outcome.
Transaction State	Planned, Validating, Authorized, Simulating, Executing, Paused, Compensating, Verifying, Completed, Failed or Awaiting Owner.
Execution Attempt	Provider, node, input, start/end, result, evidence and retry relation.
Commit Barrier	Exact irreversible transition, required evidence and authorization.
Compensation Plan	Real-world counteraction, limits, cost, authority and verification.

4. Runtime Workflow

1. Create a Planned transaction from an approved plan and bind intent, identity, authority and policy versions.
2. Validate preconditions, resources, Treaty/privacy constraints, idempotency and recovery/compensation paths.
3. Authorize execution and perform dry run or simulation according to risk policy.
4. Execute attempts with durable journal entries, safe pause/resume and bounded retries; nested transactions inherit no broader authority and cannot commit past the parent barrier.
5. Before every irreversible effect, revalidate the explicit commit barrier and required approval.
6. Verify outcomes; complete, roll back truly reversible effects, compensate irreversible effects, recover, or await Owner.

5. Interfaces and Contracts

All commands, queries and events are typed, versioned, authenticated and correlated. Commands request mutation from the authoritative owner; queries declare consistency and privacy scope; immutable events record completed facts with identity, causation, authorization, provenance and integrity metadata.

6. Failure and Recovery

- Duplicate delivery: use idempotency keys and effect records to prevent duplicate reality changes.
- Partial success: preserve completed effects and run declared compensation or Owner escalation.
- Crash during commit: reconstruct from journal and external evidence; never assume success or failure.

- Provider ambiguity: pause and verify real-world state before retry.

7. Constitutional Guarantees

- Every meaningful action has an explicit lifecycle and durable journal.
- Rollback is used only where reality is reversible; otherwise HAL compensates honestly.
- Retries never silently change original intent or broaden authority.
- Completed status requires outcome verification, not merely provider success.

8. Prohibited Behaviors

- No component may silently broaden authority, reinterpret Owner intent, erase dissent, or treat a derived projection as authoritative state.
- No degraded dependency may silently change policy, identity, trust, authentication, verification or evidence requirements.
- No provider, model, node, Presence, secret, relationship or network location receives constitutional authority by implication.

9. Observability and Verification

The subsystem records correlation, identity, policy/authority context, state versions, decisions, limitations, failures, recovery and outcomes. Verification is risk-scaled and produces durable evidence. Status reporting is derived from governed state rather than model assertion.

10. Source Alignment and Review

This chapter directly implements Decisions 9, 16, 20–22, 25, 29, 35, 40, 50, 55–56. The v0.2 rewrite restores the source-specific objects, workflows and guarantees missing from v0.1. Final disposition is recorded in the separate source-alignment and constitutional-audit reports.